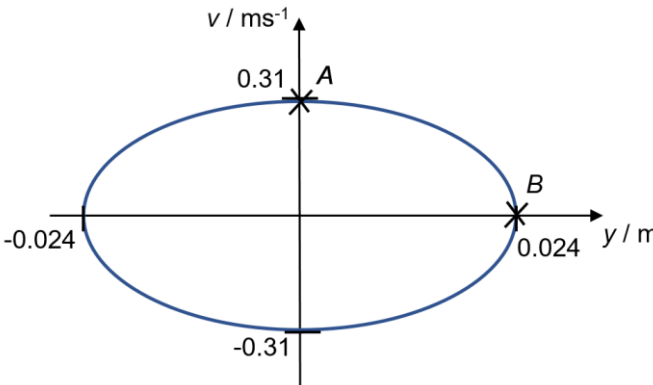
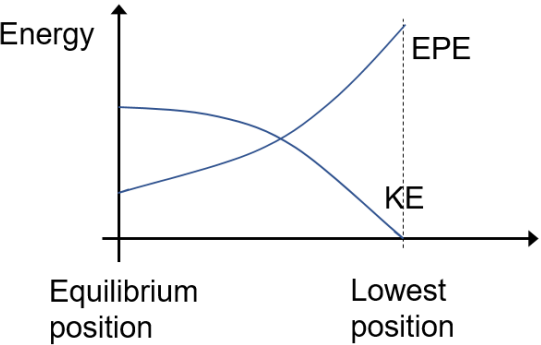


Question	Answer	Marks
3(a)	Since the acceleration of the object can be written in the form $a = \alpha y$, where α is a constant, the acceleration of the object is directly proportional to its displacement from its equilibrium point. The negative sign in the equation $a = -\frac{k}{m}y$ shows that this acceleration is always directed in the opposite direction to the displacement.	B1 B1
3b(i)	Reading off directly from the given equation, $\omega = \sqrt{\frac{k}{M}}.$ Substituting in values from the table and converting to SI units, $\omega = \sqrt{\frac{k}{m}} = \sqrt{\frac{0.25 \times 10^2}{150 \times 10^{-3}}} = 12.9099 = 12.9 \text{ rad s}^{-1}$	A1
3b(ii)	From (b)(i), we have $\omega = 12.9099 \text{ rad s}^{-1}$ Using $v_0 = \omega y_0$, we have $y_0 = \frac{v_0}{\omega} = \frac{0.31}{12.9099} = 0.0240 \text{ m}$	A1

3(b)(iii)		1m for drawing an ellipse 1m for indicating correctly the intercepts 1m for correctly indicating both points A and B.	B1 B1 B1
3(b)(iv)		Marking points: 1m for correct shape of KE graph, with maximum at equilibrium position and 0 KE at lowest position. 1m for correct shape of EPE graph, with positive EPE at equilibrium position and EPE at lowest position greater than the value of KE at the equilibrium position.	B2
	Max Marks		9

Setter: Lih Juinn Marker:			
Question	Answer	Marks	
4(a)	The e.m.f. of accumulator is the sum of p.d.s across the external resistor and the resistance wire.	A1	